



Faculty of Engineering
Mechanical Engineering
Spring - 2025/2026

Milestone 1 Report

Signals and Image Processing ENMR602

Supervised by:

Dr. Moheb Mekhaie

Eng. Moaz Fouda

Eng. Hisham Hafez

Eng. Yasmine Attia

Name	ID	Tutorial	Major
Mohamed Waleed Zoher	13006830	10	Robotics and Automation
Marwan Hesham Elqady	13006158	10	Robotics and Automation
Moaaz Bassouny	13001998	10	Robotics and Automation
Ahmed Mohamed Ramadan	13005623	10	Robotics and Automation
Youssef Mokhtar	13005346	10	Robotics and Automation

1. Introduction

This report presents the planning phase (Milestone 1) of a ball detection and classification system using classical image processing techniques. The goal is to recognise six types of balls in images: tennis ball, golf ball, football, basketball, baseball, and American football.

In this milestone, we design a feature-based pipeline that will be implemented later. The pipeline includes preprocessing, segmentation, feature extraction, and classification, with visual results at each stage.

1.1 Project Goal

Given an input RGB image containing one ball, the system should correctly classify it into one of the six categories. The pipeline is designed to be robust to variations such as lighting, background, scale, and orientation.

1.2 Class Characteristics and Feature Motivation

The six ball types differ mainly in:

- **Colour:** e.g. tennis (yellow-green), basketball (orange), football (black and white), etc.
- **Texture:** e.g. golf ball dimples, basketball surface, baseball stitching
- **Shape:** most are circular, while the American football is elongated

Based on this, the selected features are:

- **Colour histogram** → distinguishes colour differences
 - **HOG descriptor** → captures texture and local shape
 - **Shape features** → differentiate circular vs elongated objects
-

2. Dataset

The dataset of labelled ball images is provided and must be used without any modifications.

To develop the system, the dataset will be divided into three subsets:

- **Training set:** used to train the classifier
- **Validation set:** used to tune model parameters
- **Test set:** used for final evaluation

3. Pipeline Overview

The proposed system follows a sequential image processing pipeline:

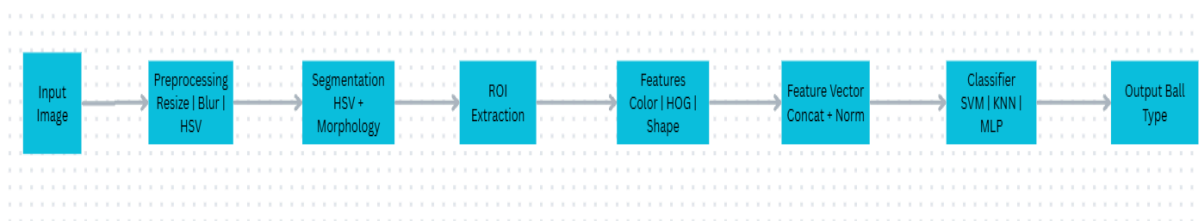


Figure 1 shows the complete block diagram of the feature-based ball classification system.

4. Preprocessing

4.1 Purpose

Preprocessing prepares the images for analysis by ensuring consistency in size, reducing noise, and converting images into a suitable color space for segmentation.

4.2 Steps:

4.2.1 Read and Resize

Images are loaded and resized to a fixed size (256×192) to ensure consistent input for feature extraction and to reduce computational cost.

4.2.2 Denoising (Gaussian Blur)

A Gaussian blur is applied to reduce noise while preserving important edges, improving the reliability of later feature extraction.

4.2.3 Contrast Enhancement

For images with poor lighting, histogram equalization is applied to the V channel to improve contrast and enhance segmentation performance.

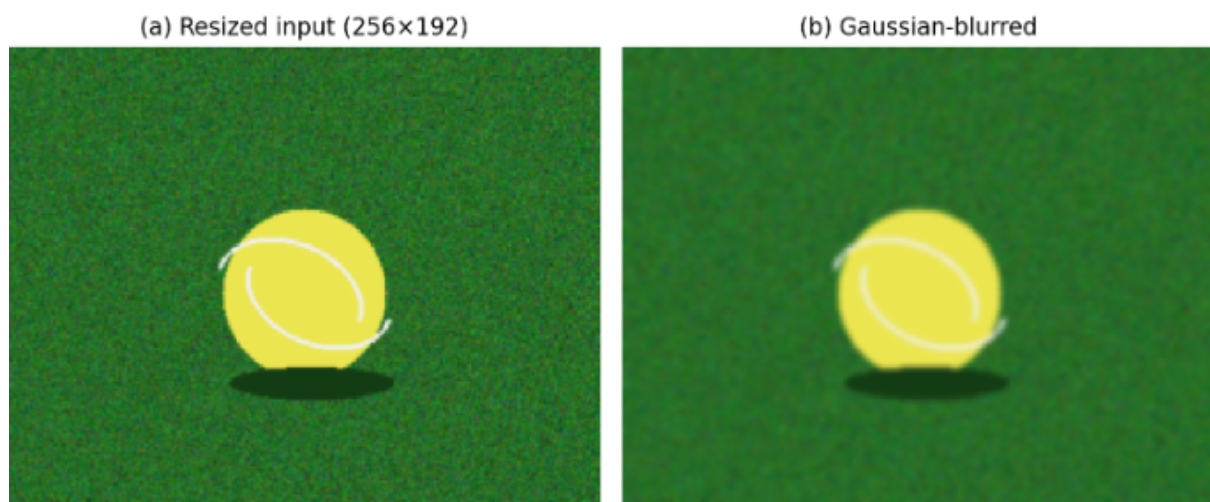


Figure 2 — (a) Resized image (256×192), (b) Gaussian blurred image.

4.2.4 Color Space Conversion (BGR → HSV)

Images are converted from BGR to HSV color space, as HSV separates color information from intensity, making segmentation more robust to lighting variations.

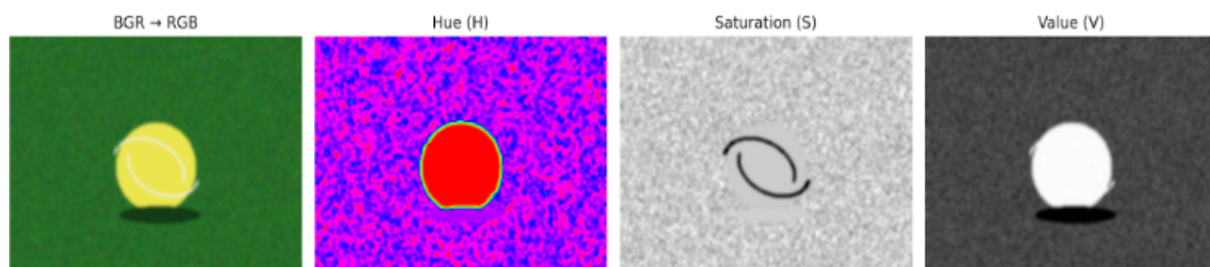


Figure 3 — HSV channels of the image. Segmentation mainly uses the H and S channels.

5. Segmentation

5.1 Objective

Segmentation isolates the ball from the background by generating a binary mask, allowing further processing to focus only on the object of interest.

5.2 Segmentation Method

A two-stage approach is used:

- **HSV color thresholding:**

The ball is detected using predefined HSV ranges for each class. This method is robust to lighting variations and efficiently isolates the ball based on its color.

- **Fallback using K-means:**

If thresholding fails (e.g. due to difficult lighting or background), K-means clustering is applied to segment the image into regions, and the most relevant region is selected.

5.3 Mask Refinement

The binary mask is refined using morphological operations:

- Opening to remove noise
- Closing to fill small holes

Finally, only the largest connected component is kept to ensure accurate object detection.



Figure 4 — Segmentation results showing the binary mask and the extracted ball region.

6. Feature Extraction

To distinguish between different ball types, three types of features are extracted: color, texture, and shape.

6.1 Color Features

A color histogram is computed in the HSV space using the segmented ball region. This captures the dominant color of each ball type and helps differentiate between classes.

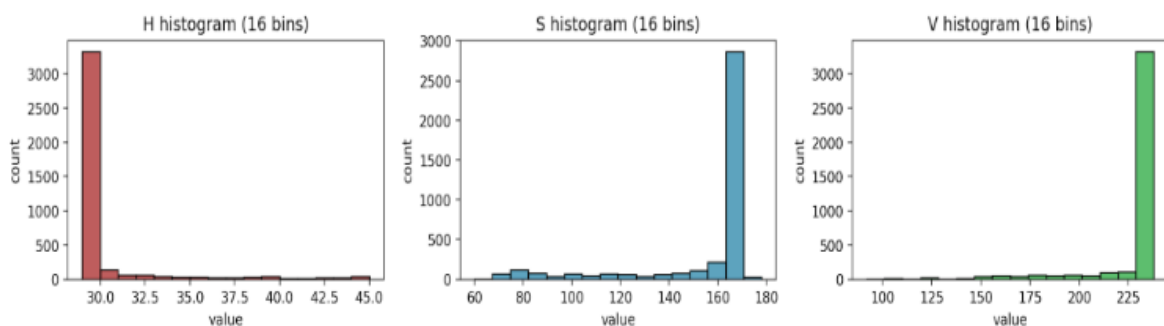


Figure 5 — Per-channel 16-bin HSV histograms computed inside the mask. Concatenating the three gives a 48-d colour feature.

6.2 Texture and Shape Features (HOG)

The Histogram of Oriented Gradients (HOG) descriptor is used to capture texture and local shape information. It helps distinguish patterns such as golf ball dimples, basketball surface texture, and football panels.

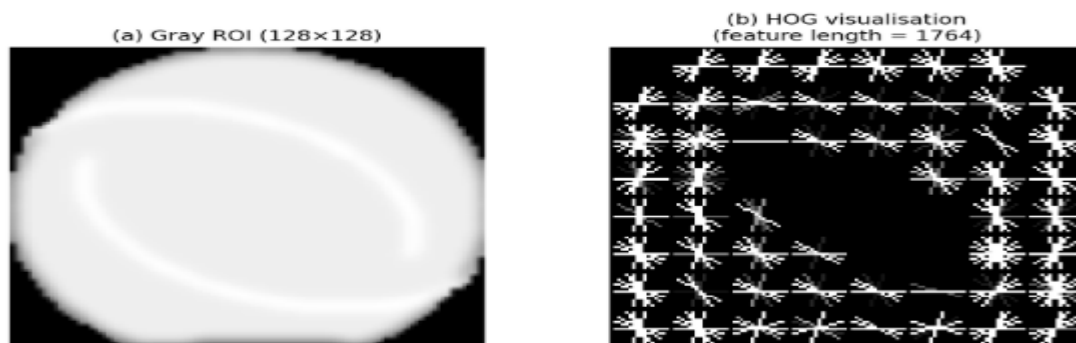


Figure 6 — HOG on the segmented ball. (a) The gray ROI cropped to the mask's bounding box and resized to 128×128 . (b) The HOG visualisation: each small star shows the dominant gradient orientations in a cell. Feature length = 1764.

6.3 Shape Features

Additional shape-based features are extracted from the segmented mask, including:

- Aspect ratio
- Circularity
- Eccentricity

These features help distinguish between circular balls and elongated shapes such as the American football.

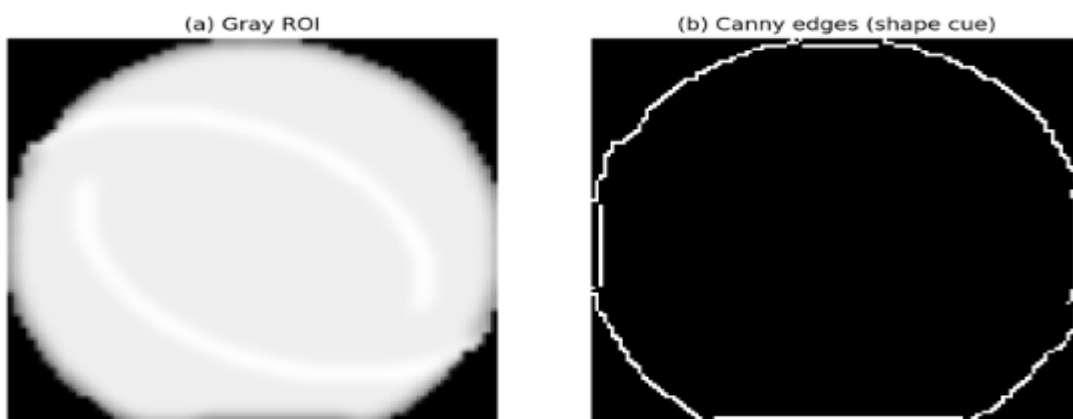


Figure 7 — Gray ROI and its Canny edges (thresholds 60 / 160). The contour-based shape statistics are derived from the closed boundary extracted from edge/mask information.

Feature Combination

All extracted features are combined into a single feature vector, which is then used for classification.

The three feature groups are concatenated into one long vector and L2-normalised as a whole:

```
x = [color_hist (48) || HOG (1764) || shape (8)]    # length = 1820  
x = x / ||x||_2                                     # L2 normalise
```

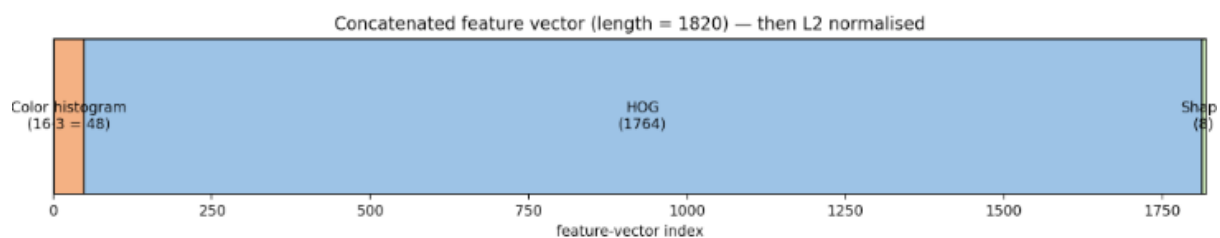


Figure 8 — Layout of the final concatenated feature vector (length ≈ 1820). Each component is itself normalised before concatenation to keep the three groups on comparable scales; an additional global L2 normalisation is then applied.

7. Classification

The extracted feature vectors are used to train a multi-class classifier that assigns each image to one of the six ball categories.

Three classifiers are considered:

- **k-Nearest Neighbour (k-NN):**
A simple distance-based method used as a baseline.
- **Support Vector Machine (SVM):**
A powerful classifier that performs well with high-dimensional features such as HOG.
- **Multi-Layer Perceptron (MLP):**
A neural-network-based approach capable of learning more complex decision boundaries.

The performance of these classifiers will be evaluated, and the best-performing model will be selected for the final system.

Classifier	Key hyper-parameters (to be tuned on val. set)	Why it is a reasonable choice
k-Nearest Neighbour	$k \in \{1, 3, 5, 7, 9\}$; distance: Euclidean / cosine	No training phase, intuitive, a natural baseline that mirrors the clustering picture in Lecture 10.
Linear & RBF SVM	$C \in \{0.1, 1, 10, 100\}$; $\gamma \in \{1e-4 \dots 1e-1\}$ for RBF	State-of-the-art on HOG features (Dalal & Triggs, CVPR 2005). One-vs-rest for six classes.
Multi-Layer Perceptron	1 hidden layer with 64 or 128 ReLU units; Adam; 100 epochs	A shallow neural baseline, reasonable on our moderate-size feature vector.

8. Evaluation Plan

The performance of the system will be evaluated using the following metrics:

- **Accuracy:**
Measures the overall percentage of correctly classified images.
- **Confusion Matrix:**
Provides insight into which classes are confused with each other.
- **Precision, Recall, and F1-score:**
Evaluate the performance for each class individually.

9. Anticipated Difficulties & Mitigations

Risk	Mitigation
White golf balls confused with other white objects.	Use texture features (HOG) to distinguish surface patterns such as dimples, stitching, or panel lines
HSV segmentation fails under difficult lighting	Apply preprocessing and use alternative segmentation methods such as clustering
Small training dataset may cause overfitting	Use validation techniques and prefer simpler models
American football appears circular in some views	Use color and texture features instead of relying only on shape

10. References

- [1] M. Mekhail, *Signal and Image Processing Lectures*, German International University, Cairo, Egypt, 2025.
- [2] N. Dalal and B. Triggs, “Histograms of oriented gradients for human detection,” in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, San Diego, CA, USA, 2005, pp. 886–893.
- [3] OpenCV, “OpenCV Documentation,” [Online]. Available: <https://docs.opencv.org>. [Accessed: Apr. 18, 2026].